

Options Market Making Project

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1 Introduction

Market makers play a critical role in financial markets by providing liquidity through continuously quoted bid and ask prices. In exchange for facilitating transactions, market makers earn the bid-ask spread but assume inventory and market risk. Managing these risks is a central challenge in quantitative trading.

The objective of this project is to build a market-making simulator from first principles and progressively introduce the risk management techniques used by professional trading firms. The project begins with a simple stock market maker and evolves into an options market-making framework incorporating inventory management, option pricing, delta hedging, Greek risk measurement, and dynamic spread adjustment.

The primary goals of the project are:

- Simulate a stochastic market environment.
- Model bid-ask market making.
- Quantify inventory risk.
- Study inventory-aware quoting strategies.
- Extend market making from stocks to options.
- Implement Black-Scholes option pricing.
- Measure Delta, Gamma, Vega, and Theta exposure.
- Evaluate the effectiveness of dynamic delta hedging.
- Investigate how risk-sensitive spreads affect profitability and risk.

2 Simulating the Market

Before constructing a market-making strategy, it is necessary to create a simulated market environment. Real market makers continuously react to changing market prices, so the first component of the project was a stochastic price process that generates a sequence of stock prices through time.

The stock price was modeled using normally distributed returns. At each time step, a random return was generated and applied to the current stock price. This allowed the stock's fair value to evolve unpredictably, producing a realistic source of market uncertainty for the simulator.

Let S_t denote the stock price at time t .

The price process was generated according to

$$S_{t+1} = S_t(1 + r_t)$$

where r_t represents a random return drawn from a normal distribution.

The purpose of this model was not to perfectly reproduce real financial markets, but rather to create a dynamic environment in which the market maker would be forced to manage inventory and risk. By introducing uncertainty into the stock price path, subsequent profitability would depend not only on spread capture but also on how effectively risk was controlled.

Figure 1 illustrates a representative stock price path generated by the simulator.

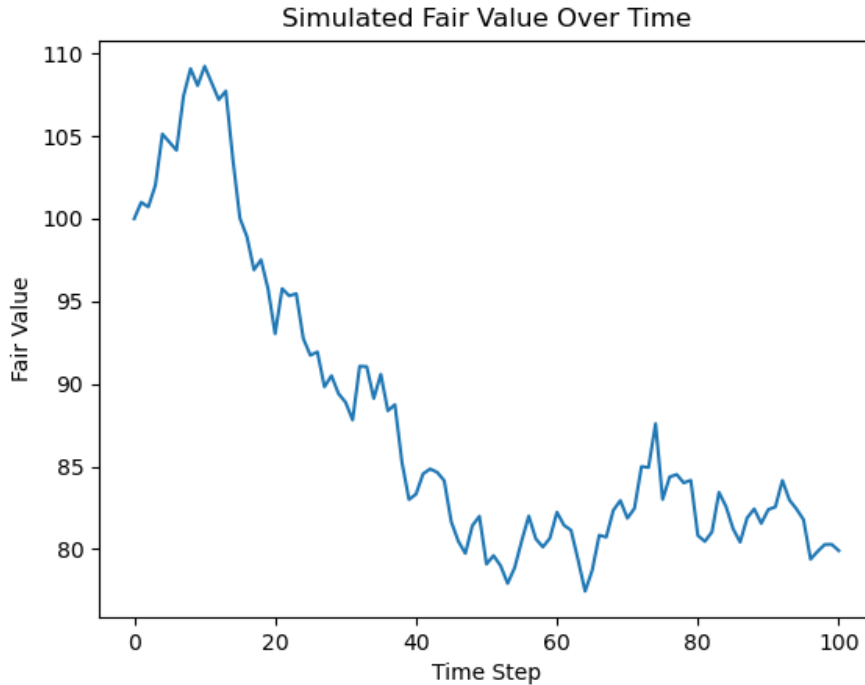


Figure 1: Simulated fair value path generated by the stochastic price process.

The simulated price path serves as the underlying market throughout the remainder of the project. All market-making decisions, inventory accumulation, hedging activity, and profit-and-loss calculations are ultimately driven by movements in this evolving fair value.

Market makers do not control the market price. Instead, they continuously react to changes in fair value while attempting to manage risk and earn profits through the bid-ask spread.

3 Basic Market Making

After constructing the simulated market environment, the next step was to build a basic market maker. The purpose of the market maker is to continuously provide liquidity by posting bid and ask prices around the current fair value.

At each time step, the market maker observes the current fair value and quotes a bid price below fair value and an ask price above fair value. Using a fixed spread, the quotes are given by

$$\text{Bid} = \text{Fair Value} - \frac{\text{Spread}}{2}$$

and

$$\text{Ask} = \text{Fair Value} + \frac{\text{Spread}}{2}.$$

The difference between the ask and bid prices is known as the bid-ask spread. This spread represents the primary source of revenue for a market maker. When a customer buys from the market maker, the trade occurs at the ask price. When a customer sells to the market maker, the trade occurs at the bid price.

In an idealized setting, repeatedly buying at the bid and selling at the ask would generate profits equal to the collected spreads. However, real market making is more complicated because customer trading activity creates inventory positions that expose the market maker to future price movements.

To model customer activity, the simulator introduces stochastic customer order flow. At each time step, a customer may:

- Buy from the market maker,
- Sell to the market maker,
- Or choose not to trade.

Each trade updates both inventory and cash balances. When a customer buys from the market maker, inventory decreases because the market maker sells shares. When a customer sells to the market maker, inventory increases because the market maker acquires shares.

As a result, inventory evolves dynamically through time and reflects the cumulative effect of customer order flow.

The simulator tracks two fundamental state variables:

- Inventory
- Cash

Inventory records the number of shares currently held by the market maker, while cash records the cumulative cash generated through trading activity.

To evaluate profitability, the simulator calculates mark-to-market profit and loss (PnL). Rather than considering only realized cash flows, mark-to-market accounting incorporates the current market value of inventory. PnL is calculated as

$$\text{PnL} = \text{Cash} + \text{Inventory} \times \text{Current Fair Value}$$

This formulation captures both realized trading profits and unrealized gains or losses resulting from inventory holdings.

Figure 2 illustrates a representative inventory path generated by the simulator.

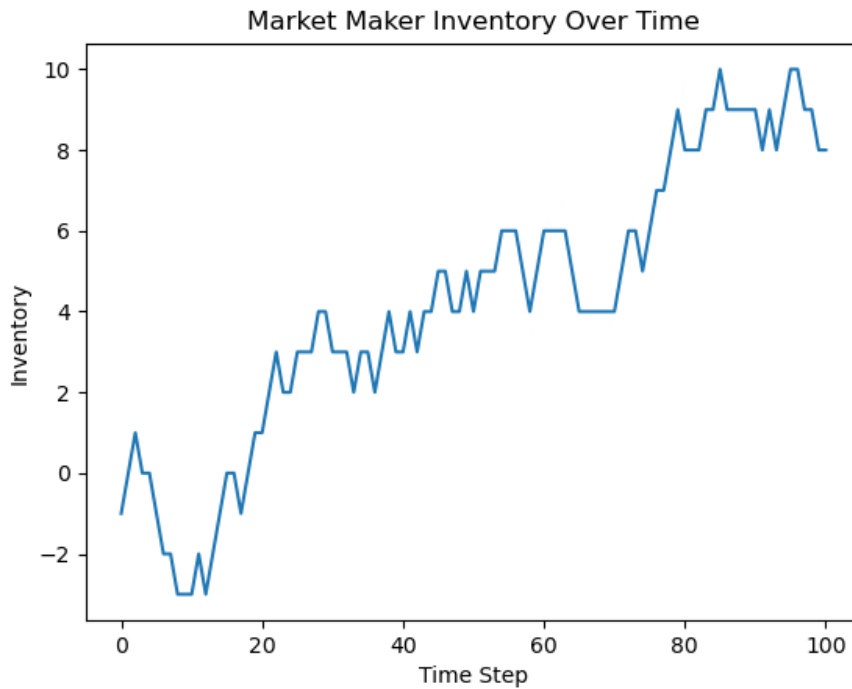


Figure 2: Market maker inventory over time.

Figure 3 shows the corresponding mark-to-market profit and loss.

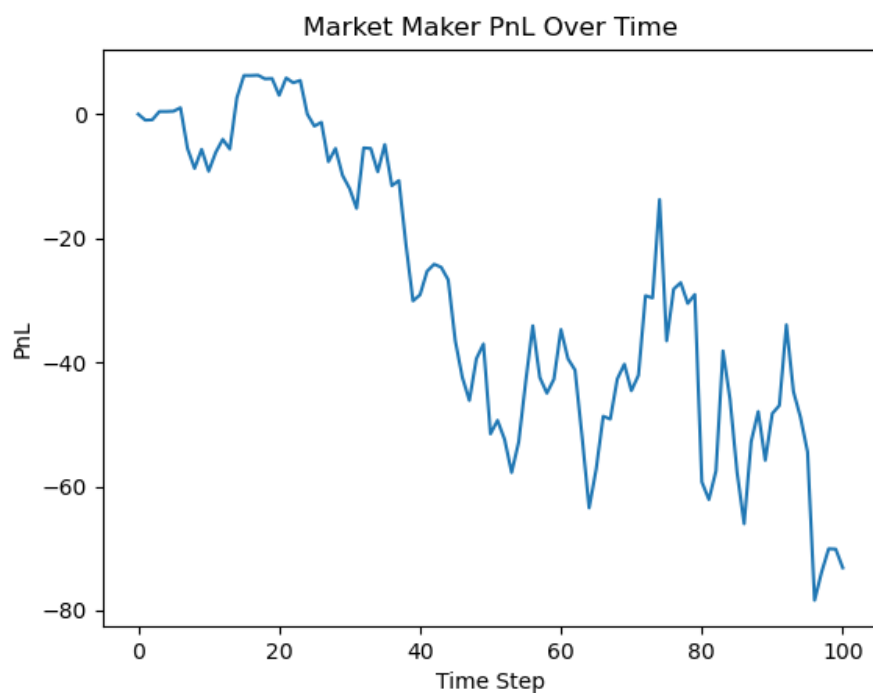


Figure 3: Market maker profit and loss over time.

These results demonstrate that profitability depends not only on spread capture but also on the inventory accumulated through customer trading activity. Even when spreads are consistently collected, adverse price movements can produce significant gains or losses through inventory exposure.

This observation motivates the next stage of the project: understanding and managing inventory risk.

4 Inventory Risk

Although the market maker earns revenue by collecting the bid-ask spread, spread capture alone does not guarantee profitability. As customer trades accumulate through time, the market maker develops inventory positions that are exposed to future price movements.

Inventory can be either positive or negative. A positive inventory indicates that the market maker owns shares of the underlying stock, while a negative inventory indicates a short position resulting from selling more shares than have been purchased.

The presence of inventory introduces directional risk. If the market maker is long inventory and prices decline, losses will occur as the value of the held shares decreases. Conversely, if the market maker is short inventory and prices rise, losses will occur because repurchasing the shares becomes more expensive.

During early simulations, situations frequently arose in which customers repeatedly purchased shares from the market maker. This caused inventory to become increasingly negative as shares were continuously sold into the market.

At first glance, this appeared beneficial because each sale generated additional cash. However, when the underlying stock price subsequently increased, the short inventory position produced mark-to-market losses that exceeded the profits earned from spread capture.

This phenomenon demonstrates one of the most important realities of market making: inventory exposure can dominate trading profits.

Figure 2 shows a representative inventory trajectory generated by the simulator. Notice that inventory can drift substantially away from zero as customer trades accumulate.

Figure 3 illustrates the corresponding profit and loss. Even though spreads continue to be collected, inventory exposure creates significant fluctuations in profitability.

The root cause of these losses is that the market maker is unintentionally expressing a directional view on the stock. A large positive inventory behaves similarly to a long investment position, while a large negative inventory behaves similarly to a short investment position.

Professional market makers generally seek to avoid large directional exposures because their objective is to earn profits from providing liquidity rather than from speculating on future price movements. Consequently, inventory management becomes a central component of market-making strategy.

The inventory risk observed in these simulations motivates the next stage of the project: inventory-aware quoting. Rather than allowing inventory to drift freely, the market maker will actively adjust quoted prices to encourage trades that reduce risk and move inventory back toward zero.

5 Inventory-Aware Quoting

After observing that inventory risk could significantly impact profitability, I introduced inventory-aware quoting. Rather than always quoting symmetrically around fair value, the market maker adjusts its quotes according to its current inventory position.

The central idea is that inventory should influence pricing decisions. If the market maker accumulates too much positive inventory, it becomes desirable to sell shares and reduce exposure. Conversely, if inventory becomes excessively negative, it becomes desirable to buy shares and reduce the short position.

To accomplish this, quoted prices are shifted according to inventory:

$$\text{Adjusted Fair Value} = \text{Fair Value} - (\text{Skew Strength}) \times (\text{Inventory}).$$

Bid and ask prices are then constructed around this adjusted fair value.

When inventory is positive, the adjusted fair value decreases. This causes both the bid and ask to move downward, encouraging customers to buy shares from the market maker and helping reduce inventory.

When inventory is negative, the adjusted fair value increases. This causes both the bid and ask to move upward, encouraging customers to sell shares to the market maker and helping reduce the short position.

The objective of this mechanism is not to eliminate inventory entirely, but rather to create a natural tendency for inventory to revert toward zero over time.

A key insight emerged during implementation. Initially, inventory-aware quoting produced almost no improvement. The reason was that customer behavior remained completely random and did not depend on quoted prices.

As a result, changing quotes had little influence on future inventory levels.

To address this issue, customer order flow was modified so that trade probabilities became sensitive to quote attractiveness. More favorable bid prices increased the probability that customers would sell to the market maker, while more favorable ask prices increased the probability that customers would buy from the market maker.

Once customer behavior responded to prices, inventory-aware quoting began to influence trading outcomes in a meaningful way.

This represented the first strategic component of the simulator. Rather than passively accepting inventory created by customer trades, the market maker could now actively manage risk through pricing decisions.

The introduction of inventory-aware quoting transformed the simulator from a simple spread-capture model into a risk-management framework capable of balancing profitability and inventory control.

6 Evaluating Inventory-Aware Quoting

After implementing inventory-aware quoting, it was necessary to evaluate whether the strategy actually improved market-making performance.

To perform a controlled comparison, the simulator was modified so that it could operate in two different modes:

- Fixed quoting
- Inventory-aware quoting

Under fixed quoting, bid and ask prices remained centered around fair value regardless of inventory levels.

Under inventory-aware quoting, prices shifted according to the current inventory position.

To reduce the impact of randomness from any individual simulation, each strategy was evaluated across 500 independent simulation runs. For each run, final profit and loss as well as inventory exposure were recorded.

The results demonstrated that inventory-aware quoting successfully reduced average inventory exposure relative to fixed quoting. However, this improvement came with a modest reduction in profitability.

This tradeoff reflects a fundamental principle of market making. Aggressively controlling inventory often requires quoting less attractive prices, which can reduce trading opportunities and lower expected spread capture.

To further investigate this relationship, I introduced a parameter called skew strength that controls how aggressively quotes respond to inventory.

A parameter sweep was performed across multiple skew strengths. For each value, the simulator was run hundreds of times and the following metrics were recorded:

- Average final profit and loss
- Average absolute inventory

Figure 4 shows the relationship between skew strength and profitability.

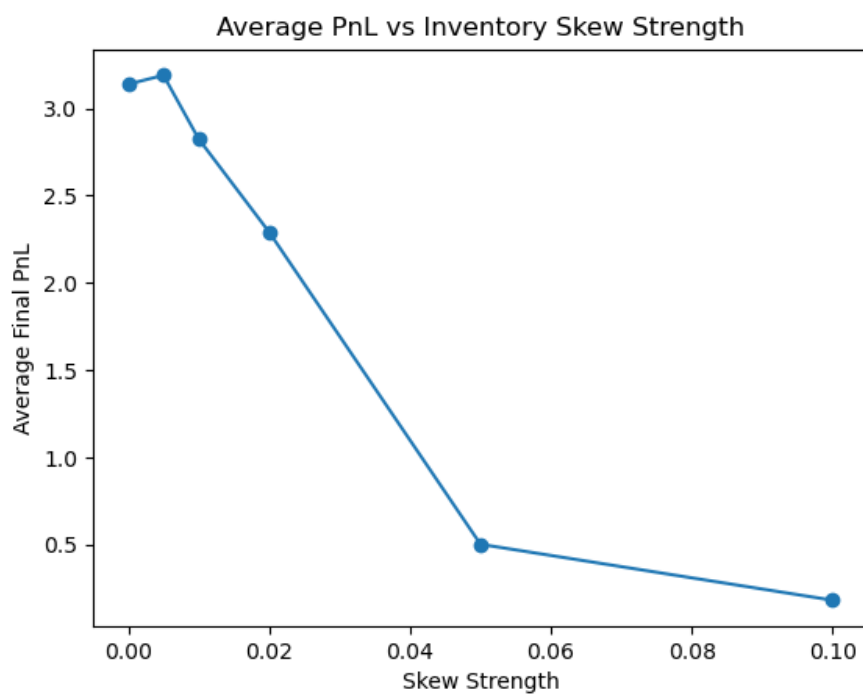


Figure 4: Average profit and loss as a function of inventory skew strength.

Figure 5 shows the relationship between skew strength and inventory exposure.

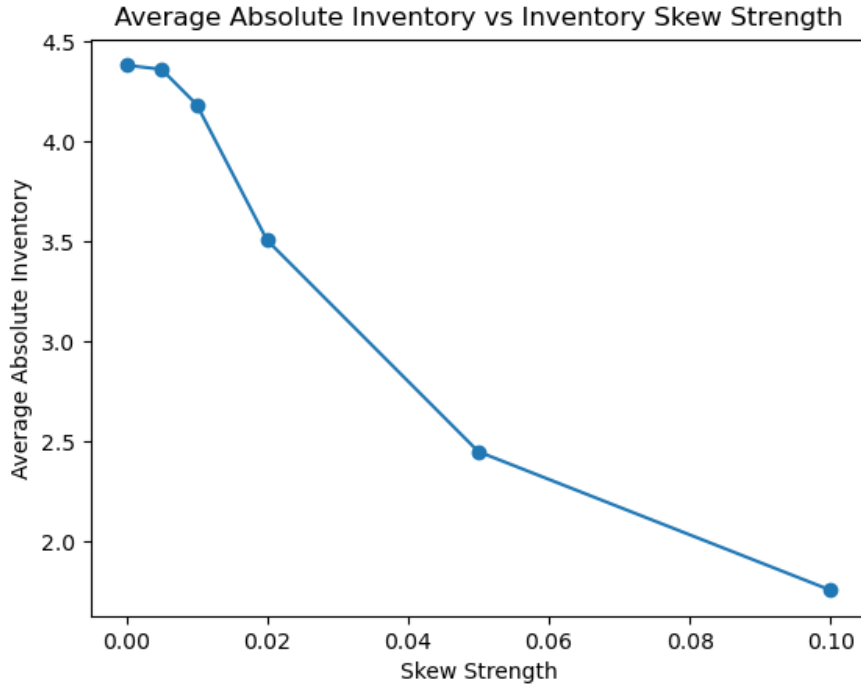


Figure 5: Average absolute inventory as a function of inventory skew strength.

The results reveal a clear tradeoff. As skew strength increases, inventory exposure decreases because the market maker more aggressively encourages inventory-reducing trades. However, profitability also declines because increasingly aggressive quote adjustments reduce opportunities for spread capture.

This experiment demonstrates that there is no universally optimal quoting strategy. Instead, market makers must balance risk management against expected profitability.

The inventory-aware quoting framework developed in this section forms the foundation for the next stage of the project, where market making is extended from stocks to options and additional forms of risk must be managed.

7 Transition to Options and Black-Scholes Pricing

While the stock market-making framework captures many important concepts, professional trading firms frequently make markets in options rather than the underlying stock itself. Options introduce additional sources of risk because their values depend not only on the current stock price but also on volatility, time to expiration, and other market variables.

To extend the simulator into an options market-making environment, I first implemented the Black-Scholes option pricing model. This model provides a theoretical fair value for a European call option based on several observable market inputs.

The Black-Scholes call price is given by

$$C = S N(d_1) - K e^{-rT} N(d_2),$$

where

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

and

$$d_2 = d_1 - \sigma\sqrt{T}.$$

The model depends on five primary inputs:

- Current stock price S
- Strike price K
- Time to expiration T
- Risk-free interest rate r
- Volatility σ

Using the stock prices generated by the stochastic price process, the simulator computes a theoretical option value at each time step. As the underlying stock price changes, the corresponding option value evolves dynamically according to the Black-Scholes model.

This creates an option fair-value path analogous to the stock fair-value path developed earlier in the project. Rather than quoting directly around stock prices, the market maker can now quote around theoretical option values.

One important distinction between stocks and options is that option values respond nonlinearly to changes in the underlying stock. A one-dollar increase in the stock price does not necessarily produce the same change in option value across different market conditions.

As a result, option market making introduces new forms of risk that are not present when trading only the underlying stock.

The Black-Scholes model provides a foundation for quantifying these risks. By calculating theoretical option values and their sensitivities to market variables, the simulator can later measure and manage option exposures using the Greeks.

This transition marks a significant increase in complexity. The stock market maker primarily managed inventory risk, whereas the option market maker must simultaneously manage inventory risk, directional exposure, volatility exposure, and time-decay effects.

The next stage of the project extends the market-making framework to options by introducing option inventory, option profit and loss tracking, and customer trading activity in the options market.

8 Option Market Making

After implementing Black-Scholes pricing, I extended the market-making framework from stocks to options.

The overall structure of the simulator remained similar to the stock market-making model. At each time step, the market maker observes the theoretical option value generated by the Black-Scholes model and quotes bid and ask prices around that value.

Using a fixed spread, option quotes are given by

$$\text{Bid} = \text{Option Fair Value} - \frac{\text{Spread}}{2}$$

and

$$\text{Ask} = \text{Option Fair Value} + \frac{\text{Spread}}{2}.$$

Customers are then allowed to interact with these quotes by either buying options from the market maker, selling options to the market maker, or choosing not to trade.

Each trade updates the market maker's option inventory and cash balance. As customer activity accumulates through time, the market maker develops a portfolio of option positions that must be managed.

Although the mechanics of option trading resemble those of stock trading, options introduce significantly greater complexity. Unlike stocks, option values are nonlinear functions of the underlying stock price and are also affected by volatility and time to expiration.

Consequently, holding option inventory creates exposures that are more difficult to manage than simple long or short stock positions.

The simulator tracks:

- Option inventory
- Cash balance
- Theoretical option value
- Mark-to-market profit and loss

Option profit and loss is calculated using the same mark-to-market framework developed earlier:

$$\text{PnL} = \text{Cash} + (\text{Option Inventory}) \times (\text{Current Option Value}).$$

This allows unrealized gains and losses from option positions to be incorporated into profitability calculations.

The resulting option market-making simulator provides a realistic environment for studying how option inventory evolves and how option risk accumulates through customer trading activity.

However, unlike the stock market maker, option inventory cannot be adequately understood using position size alone. The same inventory level can create dramatically different risks depending on the option's characteristics and market conditions.

To measure these risks more precisely, the next stage of the project introduces the option Greeks.

9 Measuring Option Risk with the Greeks

One of the most important concepts in options trading is the use of the Greeks to measure risk.

The Greeks quantify how option values respond to changes in underlying market variables. Rather than describing risk solely through inventory size, the Greeks provide a more detailed picture of how an option portfolio behaves under different market conditions.

In this project, four primary Greeks were implemented:

- Delta
- Gamma
- Vega
- Theta

Each Greek captures a different source of risk.

Delta measures the sensitivity of an option's value to changes in the underlying stock price. For a call option,

$$\Delta = \frac{\partial C}{\partial S}.$$

A Delta of 0.50 indicates that the option value is expected to increase by approximately \$0.50 when the stock price increases by \$1.

Gamma measures the sensitivity of Delta itself to changes in the stock price:

$$\Gamma = \frac{\partial^2 C}{\partial S^2}.$$

Gamma is important because Delta is not constant. As stock prices move, Delta changes, causing the portfolio's directional exposure to evolve through time.

Vega measures sensitivity to changes in volatility:

$$\text{Vega} = \frac{\partial C}{\partial \sigma}.$$

Options generally become more valuable when volatility increases because larger price movements increase the probability of profitable outcomes for option holders.

Theta measures sensitivity to the passage of time:

$$\Theta = \frac{\partial C}{\partial T}.$$

As expiration approaches, options lose time value. This phenomenon is commonly referred to as time decay.

For each option held by the market maker, the simulator calculates Delta, Gamma, Vega, and Theta using the Black-Scholes framework.

Portfolio-level Greek exposures are then computed by multiplying each option Greek by the market maker's option inventory. For example,

$$\text{Portfolio Delta} = (\text{Option Inventory}) \times (\text{Option Delta}).$$

Similar calculations are performed for Gamma, Vega, and Theta.

These portfolio-level exposures provide a quantitative description of the risks created by option inventory. Rather than merely knowing how many contracts are held, the market maker can now measure directional risk, volatility risk, time-decay effects, and the rate at which directional exposure changes.

The introduction of the Greeks transforms the simulator from a simple inventory-tracking system into a risk-management framework. These measurements form the foundation for the dynamic hedging strategies developed in the following sections.

10 Delta Hedging

Although option inventory can be measured using position size, position size alone does not accurately describe the risk of an options portfolio. The most important source of risk for an option market maker is directional exposure to movements in the underlying stock.

To quantify this exposure, the simulator calculates portfolio Delta.

For an individual option, Delta measures the sensitivity of the option value to changes in the underlying stock price. However, market makers typically hold multiple option contracts rather than a single option. Consequently, risk must be measured at the portfolio level.

Portfolio Delta is calculated as

$$\text{Portfolio Delta} = (\text{Option Inventory}) \times (\text{Option Delta}).$$

This quantity represents the approximate change in portfolio value resulting from a one-dollar change in the underlying stock price.

As customer trades accumulate, option inventory fluctuates and portfolio Delta evolves through time. Large positive or negative Delta values indicate substantial directional exposure.

To manage this risk, I implemented dynamic Delta hedging.

The central idea of Delta hedging is to trade shares of the underlying stock in order to offset option exposure. If the option portfolio has positive Delta, the market maker sells stock. If the option portfolio has negative Delta, the market maker buys stock.

The target hedge position is given by

$$\text{Stock Position} = -\text{Portfolio Delta}.$$

This causes the combined portfolio of options and stock to maintain approximately zero net Delta.

At each time step, the simulator recalculates portfolio Delta and updates the hedge position accordingly. This process creates a dynamically hedged portfolio that continuously adapts to changing market conditions.

To evaluate the effectiveness of Delta hedging, I compared hedged and unhedged option market-making strategies across 500 independent simulations.

The results are summarized in Table 1.

Strategy	Average PnL	PnL Std. Dev.	Average Absolute Delta
Unhedged	0.89	41.98	2.37
Delta Hedged	1.00	13.14	0.00

Table 1: Comparison of hedged and unhedged option market-making strategies.

The results demonstrate that Delta hedging dramatically reduced portfolio risk. The average absolute Delta decreased from approximately 2.37 to essentially zero, indicating that directional exposure was successfully neutralized.

More importantly, the standard deviation of profit and loss decreased from approximately 42 to 13. This represents a substantial reduction in risk while preserving nearly the same average profitability.

These findings illustrate one of the fundamental principles of modern options market making. Rather than speculating on the future direction of the market, professional market makers typically seek to earn profits from spread capture while using hedging techniques to minimize unwanted directional exposure.

Although Delta hedging removes most first-order directional risk, it does not eliminate all sources of risk. As the stock price changes, Delta itself changes, causing the portfolio's risk profile to evolve through time. This motivates the study of higher-order Greeks.

11 Gamma, Vega, and Theta Risk

While Delta is often the most important risk measure for an options portfolio, it is not sufficient on its own. A Delta-neutral portfolio can still experience significant gains or losses due to changes in volatility, the passage of time, or changes in Delta itself.

To capture these additional sources of risk, the simulator tracks Gamma, Vega, and Theta at both the individual option and portfolio levels.

Gamma measures the sensitivity of Delta to changes in the underlying stock price. For a call option,

$$\Gamma = \frac{\partial^2 C}{\partial S^2}.$$

Gamma is particularly important because it determines how quickly directional exposure changes as the market moves.

The simulator computes portfolio Gamma by multiplying option Gamma by option inventory. Figure 6 illustrates portfolio Gamma throughout a representative simulation.

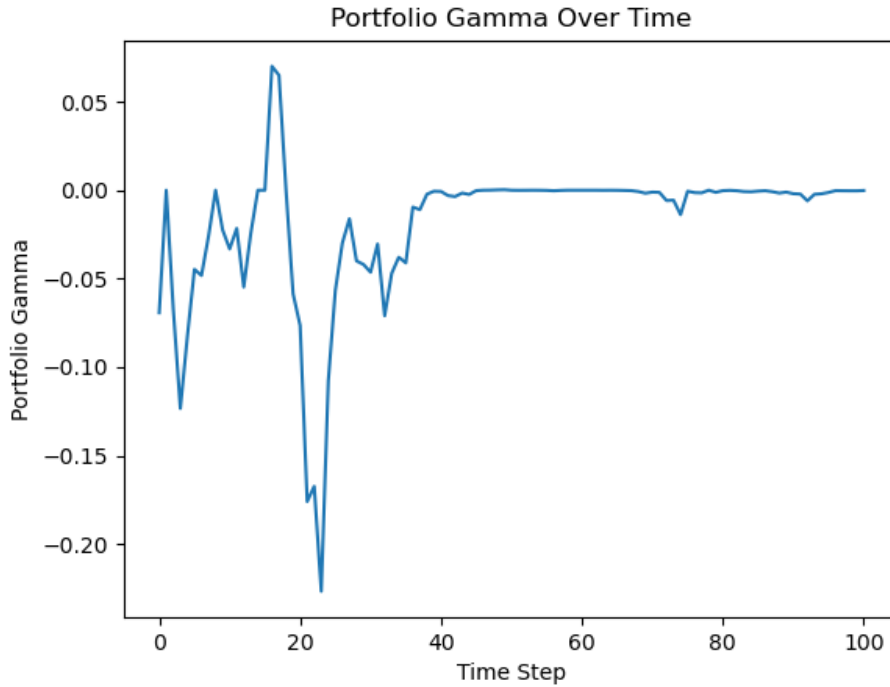


Figure 6: Portfolio Gamma over time.

The results show that even when Delta is continuously hedged, Gamma exposure remains present. This means that future stock movements can cause Delta to change, requiring additional hedge adjustments.

Vega measures sensitivity to changes in volatility:

$$\text{Vega} = \frac{\partial C}{\partial \sigma}.$$

Options generally become more valuable as volatility increases because larger price fluctuations increase the probability of favorable outcomes for option holders.

Portfolio Vega is calculated by multiplying option Vega by option inventory.

Figure 7 shows portfolio Vega during the simulation.

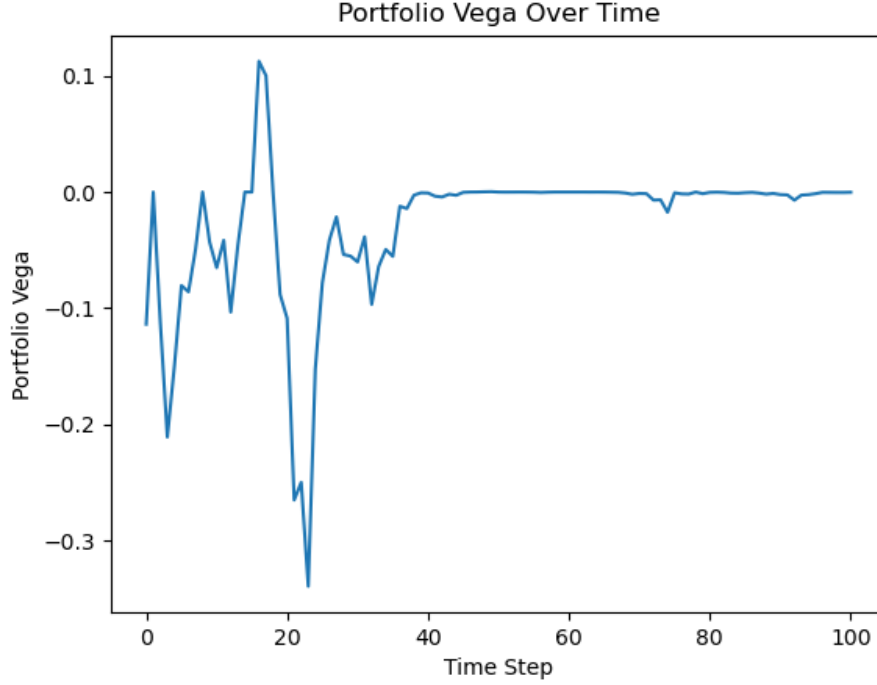


Figure 7: Portfolio Vega over time.

In many simulations, the market maker carried negative Vega exposure because it frequently accumulated short call positions through customer trading activity. As a result, increases in volatility tended to reduce profitability.

Theta measures sensitivity to the passage of time:

$$\Theta = \frac{\partial C}{\partial T}$$

Unlike Delta and Vega, Theta often benefits option sellers. As expiration approaches, options lose time value, creating profits for traders holding short option positions.

Portfolio Theta is computed by multiplying option Theta by option inventory.

Figure 8 illustrates portfolio Theta throughout the simulation.

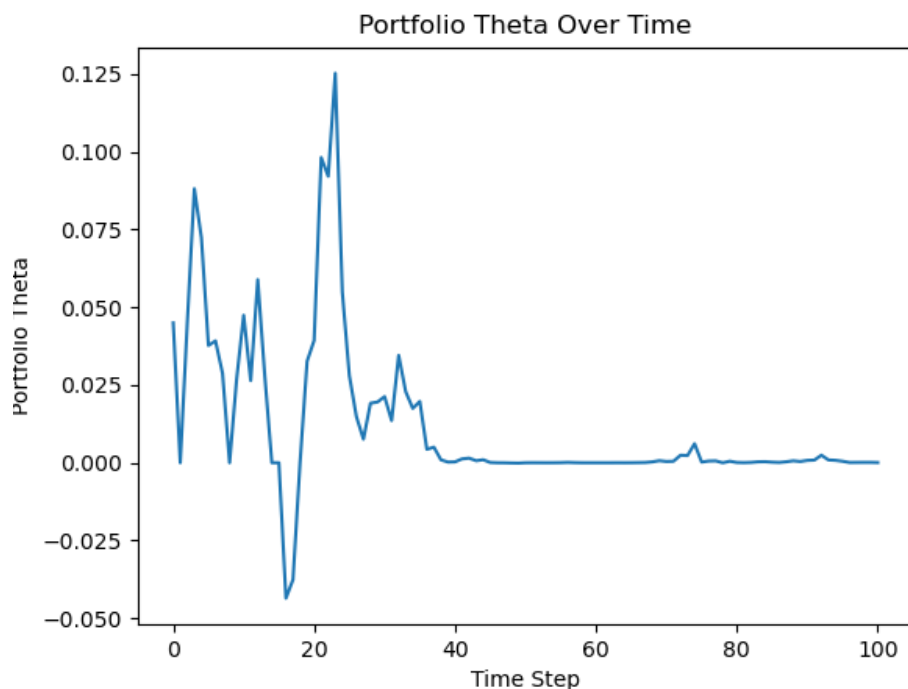


Figure 8: Portfolio Theta over time.

The simulations generally produced positive portfolio Theta because the market maker frequently held short call inventory. This reflects a common characteristic of market-making businesses: earning time-decay revenue while managing other forms of risk.

Together, Gamma, Vega, and Theta provide a richer description of option risk than Delta alone. While Delta hedging successfully removes most directional exposure, the market maker must still manage changing Delta, volatility fluctuations, and time decay.

These results demonstrate that option market making is fundamentally a multidimensional risk-management problem. Successful market makers must continuously balance several competing sources of risk while maintaining profitability.

12 Dynamic Spread Adjustment

The initial versions of the simulator used a fixed bid-ask spread regardless of market conditions. While this simplified implementation was useful for studying inventory dynamics and hedging, it does not accurately reflect how professional market makers quote prices.

In practice, market makers adjust spreads according to the level of risk associated with a position. When inventory becomes more difficult to manage or option exposures become larger, wider spreads are typically required to compensate for the additional risk.

To incorporate this idea, I implemented a dynamic spread model based on portfolio Gamma and Vega exposure.

The spread was calculated as

$$\text{Spread} = \text{Base Spread} + 2|\Gamma| + 0.5|\text{Vega}|$$

Under this framework, the spread increases when Gamma or Vega exposure becomes larger.

The intuition behind this adjustment is straightforward. Large Gamma exposure implies that portfolio Delta can change rapidly as the stock price moves, making hedging more difficult. Large Vega exposure indicates increased sensitivity to changes in volatility, creating additional uncertainty in portfolio value.

By widening spreads during periods of elevated risk, the market maker earns additional compensation for accepting these exposures.

Figure 9 illustrates the evolution of the dynamically adjusted spread throughout a representative simulation.

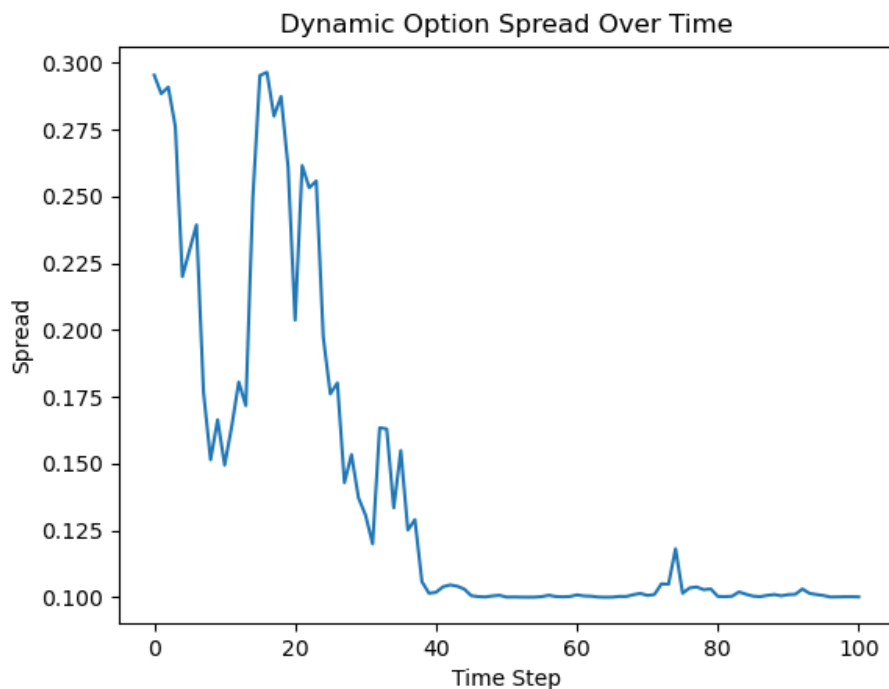


Figure 9: Dynamic option spread over time.

The results show that spreads widened when portfolio risk increased and narrowed when risk

decreased. This behavior is consistent with the practices used by many professional market-making firms.

Dynamic spread adjustment represents an important extension of the simulator because it links risk measurement directly to pricing decisions. Rather than treating risk management and quoting as separate activities, the market maker now adjusts prices in response to changing portfolio exposures.

This demonstrates how quantitative risk models can directly influence trading behavior and profitability.

13 Risk Visualization and Discussion

Throughout the project, a variety of visualizations were developed to better understand how inventory, risk exposures, and profitability evolve through time.

These visualizations provide an intuitive view of the mechanisms operating within the simulator and help validate the effectiveness of the implemented risk-management strategies.

One of the most important plots tracks total portfolio Delta through time.

Figure 10 shows the Delta of the hedged portfolio.

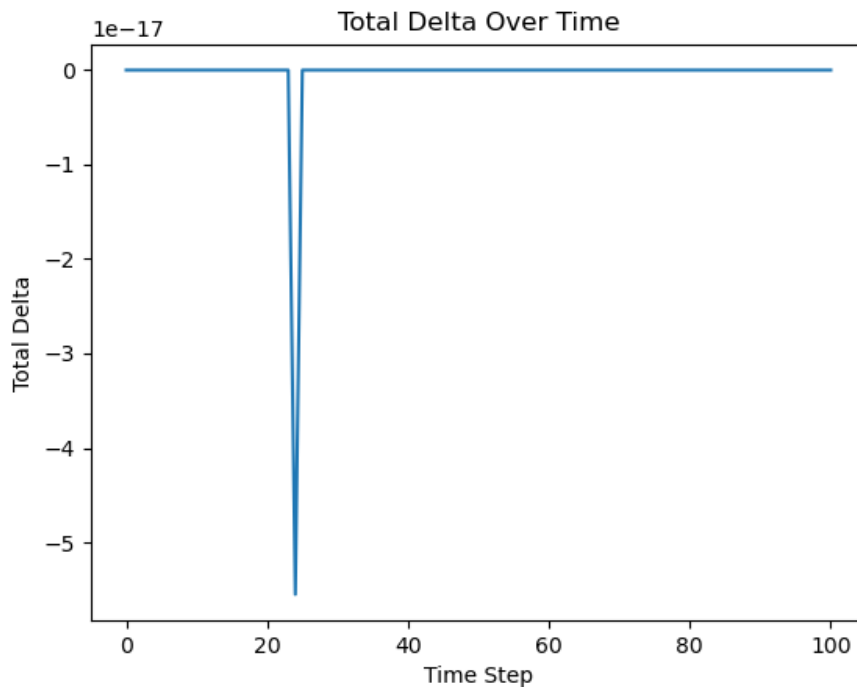


Figure 10: Total portfolio Delta over time.

The plot remains very close to zero throughout the simulation, demonstrating that the Delta-hedging algorithm successfully neutralizes directional exposure.

Figure 11 shows the stock inventory used to maintain this hedge.

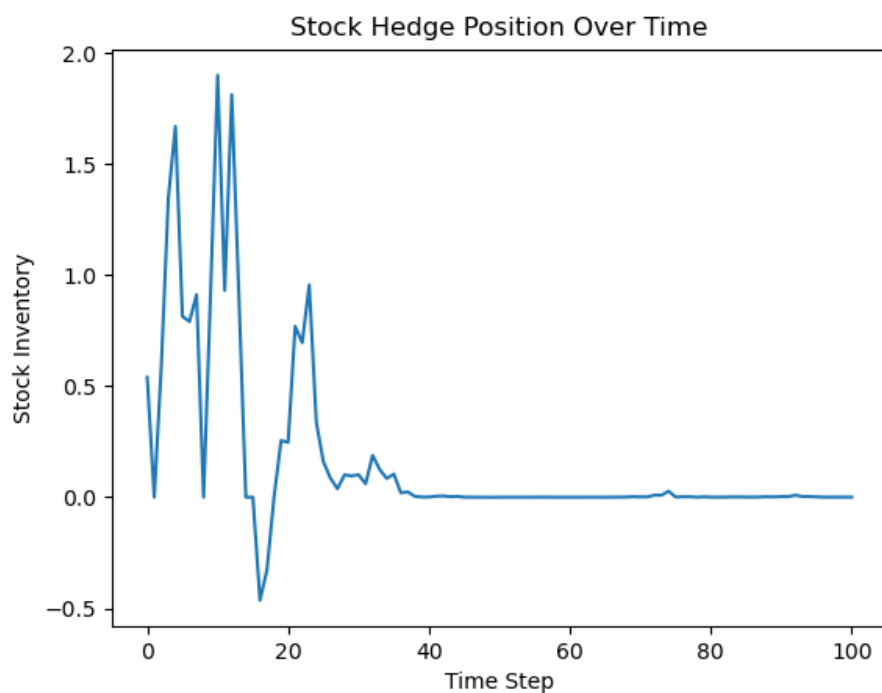


Figure 11: Stock hedge position over time.

As option inventory changes and market conditions evolve, the hedge position continuously adjusts to offset portfolio Delta. This dynamic rebalancing process is a central feature of modern options market making.

Figure 12 illustrates the profit and loss of the Delta-hedged option market maker.

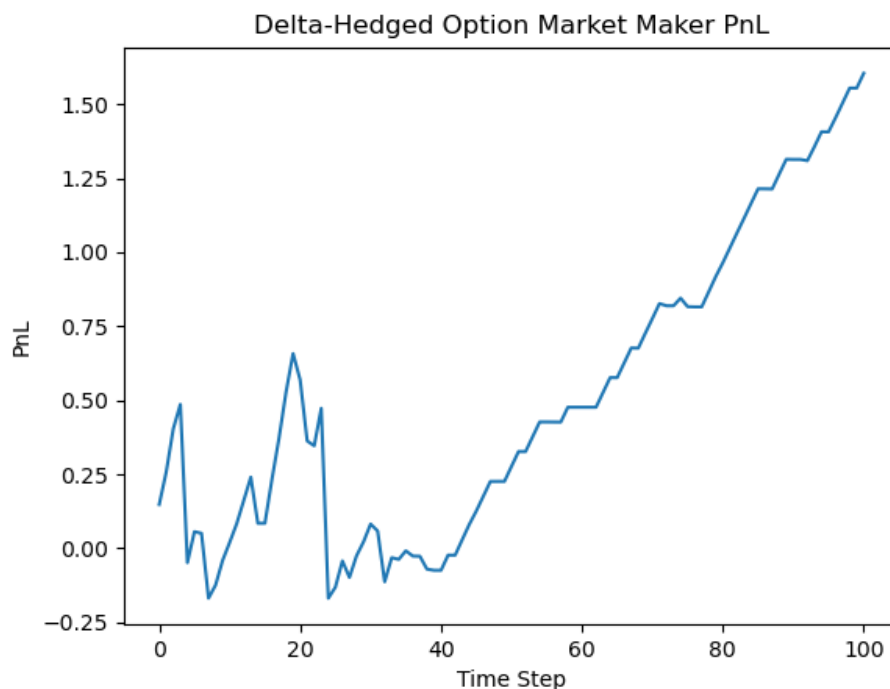


Figure 12: Profit and loss of the Delta-hedged option market maker.

Compared with earlier versions of the simulator, the Delta-hedged portfolio exhibits substantially more stable behavior. Large directional losses are reduced, allowing profitability to depend more heavily on spread capture and risk management decisions.

Taken together, the visualizations reveal several important conclusions.

First, inventory accumulation naturally creates risk that must be actively managed.

Second, Delta hedging is highly effective at removing directional exposure and reducing profit-and-loss volatility.

Third, Gamma and Vega exposures remain present even after Delta has been neutralized, illustrating that option risk is inherently multidimensional.

Finally, dynamic spread adjustment allows the market maker to respond to changing risk conditions by modifying quoted prices.

The collection of visualizations developed throughout the project provides strong evidence that the simulator behaves consistently with many of the core principles used in professional market-making environments.

14 Conclusion

The objective of this project was to build a market-making simulator from first principles and progressively introduce the core concepts used in quantitative trading and options market making. Beginning with a simple stochastic stock price process, the project evolved into a comprehensive framework incorporating inventory management, option pricing, risk measurement, dynamic hedging, and risk-adjusted quoting.

The initial stock market-making simulator demonstrated that spread capture alone is not sufficient to guarantee profitability. Customer order flow naturally creates inventory positions, and adverse price movements can generate significant losses even when spreads are consistently collected. This observation motivated the development of inventory-aware quoting, which allowed the market maker to actively manage inventory risk through pricing decisions.

Extending the simulator to options introduced additional layers of complexity. By implementing the Black-Scholes model, the simulator was able to generate theoretical option values and support option market making in a dynamic market environment. Unlike stock inventory, option inventory creates multiple forms of risk that cannot be fully described by position size alone.

To measure these risks, the simulator incorporated Delta, Gamma, Vega, and Theta calculations. These quantities provided a detailed view of directional exposure, changing directional exposure, volatility sensitivity, and time-decay effects. The resulting framework demonstrated that option risk is inherently multidimensional and requires continuous monitoring.

One of the most important results of the project was the implementation of Delta hedging. Across hundreds of simulations, Delta hedging substantially reduced portfolio risk while preserving average profitability. This finding illustrates a fundamental principle of professional market making: profitability is often driven by spread capture, while risk is controlled through systematic hedging and inventory management.

The project also demonstrated the value of dynamic spread adjustment. By widening spreads when Gamma and Vega exposures increased, the market maker was able to incorporate risk directly into its quoting decisions. This created a closer connection between risk measurement and trading behavior and provided a more realistic representation of how professional market makers operate.

Overall, the simulator successfully captures many of the central ideas underlying modern options market making. While simplified relative to real trading systems, it provides a practical framework for understanding liquidity provision, inventory management, option pricing, hedging, and risk control. More broadly, the project highlights that successful market making is not simply about predicting market movements, but rather about continuously balancing profitability and risk in an uncertain environment.